Eliza Berman

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Research Interests

My research studies bias, fairness, and stability in large language models used in high-stakes settings, such as hiring, healthcare, and scientific research. My work draws on reasoning faithfulness, uncertainty calibration, and mechanistic interpretability to efficiently measure instability in model decisions and reasoning, identify its sources, and assess how it varies across demographic groups. I aim to measure and understand bias, arbitrariness, and unfaithful reasoning in LLMs, with the goal of making LLM-assisted decisions more fair and trustworthy.

Education

New York University, Courant Institute of Mathematical Sciences — New York, NY
Ph.D. in Computer Science — August 2025 – Present

- Supported by the NSF Graduate Research Fellowship
- Advised by Professors Daniel B. Neill and Emily Black
- Focus: Fairness, bias, and stability in AI and LLMs

Brown University — Providence, RI
B.A. in Computer Science and Hispanic Studies — May 2023

- Honors: Magna Cum Laude, Phi Beta Kappa

Publications and Presentations

*Equal contribution

Peer-Reviewed Conference and Journal Papers

[1] **Eliza Berman**, Bella Chang, Daniel B. Neill*, Emily Black*. *Attribution Bias in Large Language Models.* Conference on Language Modeling (COLM), 2026. **Oral Spotlight** (top 2.8%, 24/856). [arXiv]

[2] **Eliza Berman**, Holly Sundberg Malek, Michael Bitzer, Nisar Malek, Carsten Eickhoff. *Retrieval Augmented Therapy Suggestion for Molecular Tumor Boards: Algorithmic Development and Validation Study.* Journal of Medical Internet Research (JMIR), 2025; 27:e64364. [DOI]

Workshop Papers

[3] **Eliza Berman***, Rohan Subramanian*, Shreya Jha, Lohit Jagarapu, Rajesh Ranganath. *Solver to System: LLM Reasoning Across Representations and Execution in Scientific Simulation.* Sim2Science Workshop at NeurIPS, Paris, 2026. [OpenReview]

Posters

[4] **Eliza Berman***, Alexander Chong*, Shruta Rawat, Alpana Dange, Murugesan Sivasubramanian, Vivek Anand, Viraj V. Patel. *Leveraging Natural Language Processing to Identify Priorities of Young Sexual and Gender Minority People in a Social Media Group: Implications for Health Interventions.* 15th Annual Machine Learning Symposium, New York Academy of Sciences, New York, NY, 2024.

Manuscripts in Preparation

[5] Rachel Berman, Erin Goss, Sandra Braganza, **Eliza Berman**, Iman Hassan, Zoon Naqvi, Manaslu Gurung, Emily Buttigieg, Victoria Gorski. *A Residency Curriculum to Support the Integration of Social Care into Health Care.*

Industry Experience

Microsoft Redmond, WA
Software Engineer Intern, Office Experiences June 2022 – August 2022

- Optimized big-data pipelines for Office retention and collaboration metrics; built Power BI dashboards

Audible Remote
Software Development Engineer Intern, Quality Assurance June 2021 – August 2021

- Migrated a service across frameworks using AWS tools to improve efficiency and readability; monitored service metrics

Teaching Experience

New York University, Tandon School of Engineering New York, NY
Teaching Assistant, CS-GY 9223: Responsible AI and the Law Fall 2026

The Open Door New York, NY
English as a Second Language Teacher, Adult Night School January 2025 – Present

The Fulbright Program Peru
English Teaching Assistant March 2024 – December 2024

Brown University, Department of Computer Science Providence, RI
Head Teaching Assistant, CSCI 1430: Computer Vision Fall 2022
Teaching Assistant, CSCI 1430: Computer Vision Fall 2021
Teaching Assistant, CSCI 0220: Discrete Structures and Probability Spring 2021

Leadership and Service

NYU PhD Women in Computer Science New York, NY
Board Member 2025 – Present

Women in Computer Science (WiCS) at Brown Providence, RI
Coordinator May 2020 – May 2023

Brown Meiklejohn Peer Advising Providence, RI
Peer Advisor to First-Year Students August 2020 – May 2022

Awards and Honors

- **NSF Graduate Research Fellowship** (2026)
- **Fulbright U.S. Student Program, English Teaching Assistant Award, Peru** (2024): One of six U.S. citizens selected
- **Fulbright Peru Projects Grant** (2024): Selected from all Fulbright grantees in Peru to receive funding to expand a project
- **Barbara Banks Brodsky Prize for Excellence in Real World Writing** (2023): Best Essay Award
- **Brown University Grace Hopper Celebration Scholarship** (2021, 2022): Funded attendance at the Grace Hopper Celebration of Women in Computing